Deep Reinforcement Learning for Boosting Individual and Aggregate Diversity in Product Recommendation Systems

Jijun Yu*
Shanwei Institute of Technology
Shanwei, Guangdong, China
13530591820@163.com

Shuojun Lyu
Shanwei Institute of Technology
Shanwei, Guangdong, China
lyu.00@qq.com

Peihui Chen
Shanwei Institute of Technology
Shanwei, Guangdong, China
cph2eddie@163.com

Abstract

The significance of individual and aggregate diversities has been recognized as an important aspect of the recommendation algorithm. Traditional diversity-enhancing recommendation systems have predominantly optimized accuracy and diversity through static rating processes, neglecting the dynamic, long-term user-recommender interactions. This study proposes a novel approach employing deep reinforcement learning (DRL) to derive an optimal selection policy that enhances accuracy and individual and aggregate diversities through maximizing expected cumulative rewards. This approach employs Deep Reinforcement Learning (DRL) to create a dynamic recommendation list that integrates diversity using a Markov decision process based on user feedback and ratings. It calculates the rewards by examining the proximity of the recommended list to an ideal distribution across all recommendations, thereby promoting both individual and collective diversity in the items selected. The learning mechanism is powered by the Deep Deterministic Policy Gradient (DDPG) method to refine the selection strategy. Comprehensive tests on datasets from Alibaba and Grouplens confirm that our method achieves accuracy and substantially improves diversity compared to baselines.

CCS Concepts

• **Information systems** → Information systems applications; Enterprise information systems.

Keywords

Recommender systems, Individual diversity, Aggregate diversity, Reinforcement learning

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*Corresponding author.

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1 INTRODUCTION

Recommender mechanisms show a crucial part in e-commerce by aiding customers in locating products. Research indicates that these systems can boost e-commerce sales [1-2]. The performance of recommender systems is typically assessed based on how accurately and diversely they can provide recommendations [3-6], specifically in terms of matching a customer's product preferences. Historically, the focus has been on enhancing the accuracy of recommendations, as evidenced by numerous studies aimed at refining these methods. However, multiplicity in recommendations has also emerged as a important for enhancing customer satisfaction and increasing sales within e-commerce [4, 7-10].

Prior studies identify two primary forms of diversity within recommendation systems: individual diversity (ID) focuses on the variety of items within a single customer's recommendation list, aiming to satiate distinct customer needs and enhance the unexpected discovery of products, thereby boosting customer satisfaction [11-12]. Next, the aggregate diversity (AD) measures the variation in recommended products across all customers. A lack of AD suggests a narrow range of products being frequently recommended, contributing to the long tail effect observed in e-commerce. Enhancing AD by promoting less common products can significantly boost profitability, given the vast availability of these less popular items. Thus, AD holds considerable importance for e-commerce platforms [13-16]. While both accuracy and diversity are pivotal, balancing them is challenging; typically, enhancing diversity reduces accuracy. Moreover, improving ID does not inherently increase AD; the reverse is also true [4, 16]. For instance, a high ID could be achieved by recommending a varied set of products universally, which does not necessarily enhance AD. Various strategies like re-ranking [17] and optimization-based approaches [18-21] have been developed to balance accuracy with either form of diversity. However, most efforts have been isolated to address the balance between accuracy and one type of diversity at a time. Limited research has been conducted on balancing all three aspects simultaneously, such as the studies by Wang Q et al., which explored overall balance through re-ranking methods without considering ongoing user-item interactions [22-23].

In the context of online users' choice of products, recommendation lists can also present the problem of choosing similar or two types of diverse items (ID or AD) to show to the user given her current state and given the current context. More specifically, the recommender agent has three types of interactions with the users: first, interactions in which similar products are generated, which we denote as accuracy-oriented recommendations, and second, interactions in which individual diverse products are generated based on the distance of the recommended products, which we denote as

individual diversity-oriented recommendations. Third, the interactions for generating aggregate diverse products are based on the distribution of recommended products to all customers, which we denote as aggregate diversity-oriented recommendations. Drawing on insights from existing literature on recommendation diversity, predominantly focused on single diversity measures, we frame the recommendation process as a multistep sequential ranking challenge. In response, we introduce an innovative DRL algorithm aimed at devising an optimal ranking policy for diversity in endorsements. The key innovations of this research work are outlined below.

This paper reinterprets recommendation sessions focusing on individual and aggregate diversity as Markov decision processes, incorporating three critical aspects—accuracy, individual diversity, and aggregate diversity. This approach helps structure the diverse ranking challenge by defining specific state spaces, reward functions, and transition functions. We introduce a novel interactive diversity recommendation algorithm utilizing deep reinforcement learning, designed to balance relevancy with both types of diversity in the recommendations provided. Our approach extends beyond the traditional single diversity focus by calculating distances in a sole customer's suggested items and across all customer suggestions. This strategy aims to mitigate the long-tail effect in product distribution and elevate user satisfaction, fostering a beneficial outcome for both users and platforms.

To streamline the recommendation process, our method involves the actor network identifying similar items to a proto-action, from which it selects the top-N items that score highest in terms of individual and aggregate diversity. We validate our algorithm through testing on the MovieLens and Alibaba datasets, where it significantly outperforms existing baseline methods.

The organization of this paper is as follows: Section Two delves into recommendation techniques that prioritize diversity and accuracy, along with discussions on upholding a balance between these elements. Section Three describes our proposed methodology. Section Four discusses the experimental outcomes, and the final section concludes the study, highlighting key findings and prospective areas for further research.

2 BACKGROUND

This section outlines the fundamental concepts of diversity in recommendation systems and deep reinforcement learning (DRL). We explore how DRL can be adopted to improve the proficiency of diversity-focused recommendation methods.

2.1 Recommendation methods for the accuracy-diversity trade-off

Diversity is recognized as a crucial metric in assessing recommendation systems, prompting the development of various methods to enhance this aspect. Literature reveals two primary approaches for integrating diversity into recommendations. Initially, some works focus on incorporating diversity early in the recommendation pipeline, specifically during the preference estimation phase. For instance, Niemann et al. [24] enhanced the accuracy of niche product ratings by accounting for their joint consumption in similarity measurements. Similarly, Gan et al. [3, 25] addressed the negative impacts of item popularity by modifying user similarity calculations with a power function, aiming to refine both rating predictions and overall recommendation quality. Additionally, Gan et al. [3] employed a random walk with restart approach on a user similarity network to assign discriminant scores to various products, thereby tackling diversity deficits directly at the source. These strategies, while effective at introducing diversity at the outset, tend to maintain static levels of accuracy and diversity throughout recommendation.

Conversely, the second approach maintains diversity throughout the product selection phase. This method involves using established accuracy-focused techniques to predict customer preferences and then selecting items based on these predictions alongside criteria that prioritize diversity. For example, some works [12, 18, 25] approached the accuracy-diversity dilemma as a multi-objective optimization challenge, setting specific constraints to find feasible solutions, albeit often struggling to pinpoint the optimal resolution [26-27]. Others have introduced new ranking methods intended to produce more balanced recommendations, for example the popularity-weighted ranking method with parameterization and the re-ranking strategy using graph [14]. Despite these efforts, most previous methods primarily sought a balanced compromise between accuracy and various diversity dimensions, often overlooking the simultaneous optimization of all three critical factors. Recently, Wang et al. [25] and Çanakoğlu et al. [22] devised a optimization method based on multiple objectives that employs re-ranking to fine-tune the balance between accuracy and diversity. However, these models treat the recommendation process as static, disregarding the ongoing interactions typical of an online, real-time setting. This paper proposes a novel approach by employing DRL to conceptualize diversity in recommendations as a dynamic process, thereby addressing this gap.

2.2 Recommendation methods based on RL

Reinforcement learning (RL) is an approach where an intelligent system iteratively acquires optimal manners over trial-and-error exchanges with its surroundings, fundamentally grounded in the Markov decision process framework [28, 29]. Various scholars have applied this framework to address complex decision-making sequences for recommender systems. Shani et al. [6] uses a Markov decision process together with n-gram approach, leading the way in framing recommender systems as sequential decision-making tasks. Concurrently, Zhao et al. [30] built a deep reinforcement learning-based listwise model (LIRD) to enhance interactive recommendation processes. Hu et al. [29] addressed the ranking control issue in e-commerce by applying a deterministic policy gradient method within a session-based MDP framework. Similarly, Zheng et al. [31] coined a Deep Q Network (DQN) method tailored for custom-made news suggestion, focusing on long-term reward maximization and strategic exploration to diversify recommendations. Liu et al. [32] designed a novel framework, DRR, for deep reinforcement learning that intensively models user-item interactions. o manage large sets of candidate products efficiently, Chen et al. [33, 34] introduced a cascade DQN approach. Duan et al. [35-38] applied reinforcement learning for multi-objective optimization in natural

language processing (NLP) to select the most suitable words based on diversity and accuracy.

The application of RL in recommendation engine treats the recommendation process as a dynamic, interactive sequence of decisions [39]. These methods offer three significant advantages: they adaptively update recommendations based on user feedback, balance between exploiting known user preferences and exploring new options to capture evolving interests, and aim to maximize long-term rewards, thus aligning with users' prolonged satisfaction. RL in recommendation systems is broadly categorized into Multi-Armed Bandit (MAB) and Markov Decision Process (MDP)-based approaches. MAB-based methods view the problem as a Contextual Multi-Armed Bandit issue, incorporating user and item features as context. Meanwhile, MDP-based strategies, which include both model-based and model-free variants, focus on optimizing both immediate and future returns. Model-based methods utilize transition probabilities to estimate optimal policies, but these are often challenging to determine accurately in practical scenarios, limiting their application. In contrast, model-free methods assume unknown transition probabilities, allowing the recommendation agent to learn from historical user interaction data. Several reseacrhers explored the use of partially observable MDPs in commercial settings, although scalability remains a challenge due to the exponential growth in state space with larger item sets. Advances in deep learning have mitigated these challenges by efficiently processing large-scale data and complex patterns. Additionally, the DQN framework, such as DRN for online news recommendation [40], and the usage of methods like Deep Deterministic Policy Gradient in news recommendation scenarios [41] have shown promise in managing extensive action spaces and improving recommendation precision.

While many existing methods in reinforcement learning-based recommendation systems focus primarily on optimizing accuracy, there is a growing recognition of the importance of diversifying recommendations. These approaches underscore the necessity of considering both individual diversity (ID) and aggregate diversity (AD) alongside accuracy in recommendation systems. In dynamic environments, recommender systems continuously adapt by estimating users' diverse preferences and recommending corresponding items. Let U denote the user set and I the item set. For each user $u \in U$, the system generates a sequence of recommendations $A = [a_1, a_2, ..., a_T]$ with item $i \in I$. User interactions are defined as decisions made in response to these recommendations. Feedback from these interactions—such as ratings or transaction logs—guides the ongoing adjustment of the recommendation policy π., which will be updated systematically based on the feedback $o \in O$ received during the interaction at the time stamp t. Within an environment encompassing all available items i, user interactions initiate from an initial state s derived from their operational log feedback, prompting the generation of a tailored recommendation list. User responses to this list L, typically characterized as purchase decisions or non-purchase indicators (such as ratings, browsing, buying, or clicking actions), inform subsequent updates. The agent will employ user feedback to refine its recommendation strategies before progressing to subsequent interaction episodes. This paper treats the process of recommending diverse items as a series of sequential communications among handlers and the recommendation engine. The system is designed to autonomously determine the most effective recommendation tactics by utilizing reinforcement learning.

RL in recommendation systems is generally characterized into value and policy-based strategies. Actor-critic networks, which represent a hybrid of these two approaches, combine the robustness of Q-learning with the flexibility of policy gradients. These networks consist of an actor, which optimizes actions under policy π, and a critic, which assesses the policy's effectiveness. This dual mechanism facilitates precise action selection in complex scenarios. Given the benefits of actor-critic methods over traditional Deep Q-Network (DQN) algorithms, this paper advocates for their use in developing models that not only consider recommendation accuracy but also enhance both individual and aggregate diversity.

3 METHOD INTRODUCTION

This section outlines our developed method aimed at enhancing both individual and aggregate diversities in product recommendations (dubbed D^3RDRL). Initially, we will provide definitions for key terms and describe the specific challenges that recommender systems face within the RL context. Subsequently, we introduce an RL framework based on the actor-critic approach. We will then explain the process of training this framework using data from user behavior logs and its application in making recommendations that prioritize both individual and aggregate diversity. Detailed explanations of each aspect are presented in the subsequent subsections.

3.1 Basic definitions

Within the interactive setting of the diversity recommendation system, users are presented with a curated list of items that balance both similarity and distinctiveness. Users engage by selecting an item from this list. This sequence of actions and interactions is repeated over time to optimize the total cumulative reward of the system, and this process is structured as a MDP. We outline the MDP's Markov process as follows:

State space: a state s_t encapsulates the embedding of a handler's new positive interactions prior to the time step t, alongside the user's individual characteristics. At this point τ, the recommender system selects an action according to the strategy π_τ. This action transitions the system from its present state to a new state $s_{\tau+1}$. The system then pauses, awaiting the user's subsequent interaction.

Action Space: Action a_t represented as a discrete parameter vector, is crucial for shaping the recommendation list to include either similar, interdiverse, or intradiverse items, as suggested in [32]. the action can be denoted as $a_t \in I_{1*K}$. Every item $i_t \in I_{1*K}$ possess a place score, which is referred as the inner product of the item embedding and action, i.e.,$i_t a^T$. Then, the top-rated choices are recommended. Specifically, the embedding for each item, denoted as i_t, can be derived using algorithms focused on optimizing accuracy, individual diversity, or aggregate diversity.

Action Selection Strategy π: The method for choosing which items to recommend is defined by the selection strategy employed by the recommender system. In this study, we apply the same selection approach as in DQN, which utilizes a greedy policy to select recommendation items.

Action Reward r: Following the selection of an action a_t in its current state s_t, the recommendation engine presents the handler

with a selection of products that are both varied and similar. The handler associate with these products and delivers comment such as ratings, clicks, or purchases. In response, the system captures immediate rewards r and updates its policy to enhance its capability to offer a list that is diverse in terms of both ID and AD. As referenced in [26], Borda's rank aggregation method has been proven to enhance the efficiency of recommendation engine by optimizing for accuracy, ID, and AD. Drawing on these findings, we employ the following scoring function:

$$\text{Score}_{(i,t)} = \alpha \times \text{Ranking}_{(i,t)}^{\text{List}_{\text{accuracy}}} + \beta \times \text{Ranking}_{(i,t)}^{\text{List}_{\text{ID}}} + \gamma \times \text{Ranking}_{(i,t)}^{\text{List}_{\text{AD}}} \tag{1}$$

where $Ranking_i^{List}$ is the ranking of item i in the related ranking order, α, β, and γ are utility allocated to respective ranking lists, $0 \leq \alpha, \beta, \gamma \leq 1$ and $\alpha + \beta + \gamma = 1$. By integrating these three weighted ranking lists, the composite recommendation list is derived as follows:

$$Recommendation_list = \left\{ I_1^{'''}, I_2^{'''}, \ldots, I_N^{'''} \right\}$$

where $Score(I_i^{'''}) \leq Score(I_{i+1}^{'''})$.

Finally, the action a_t is executed by the recommendation_list. At timestep t, an intelligent system presents an item from *Recommendation_list* to the user by $\text{rewardU}_{(i,t)}$. If user i actively rates item j, this rating is used; if not, a predicted rating generated by the recommender is employed instead. Based on this interaction, the reward function is formulated as follows:

$$\text{RewardU}_{(i,t)} = \begin{cases} 1, & if\ the\ item\ comes\ from\ the\ Recommendation_list \\ 0, & else \end{cases} \tag{2}$$

Transition probability: Upon executing the action a_t at a specific time step t, the recommender system transitions from state s_t to s_{t+1}. If the user disregards the recommendations, the state does not change, remaining as. Conversely, if the user engages with the recommendations, the state evolves based on this feedback. The transition probability, $p(S_t|S_{t+1}, a)$, represents the likelihood that the handler's state transitions from S_t to S_{t+1} following the deed taken by the recommender. This probability $p(S_t|S_{t+1}, a)$ is set to 1 if the handler relates with the suggested product, indicating a definite change in state; if no interaction occurs, the transition probability is 0.

The discount rate γ, can either be 0 or 1. This parameter scales the value of upcoming recompenses compared to instantaneous rewards. A $\gamma = 0$makes the agent focus solely on immediate rewards, exhibiting short-sighted behavior. Conversely, a $\gamma = 1$indicates a preference for future rewards by the agent. Therefore, the reward expected over time, discounted at each step, is indicated by G_t. In more detail, our focus is on devising a method for generating that optimizes for accuracy and ID and AD. The steps to achieve these outcomes are outlined based on the methodologies described in Ref. [23].

3.1.1 Accuracy-oriented ranking list. Typically, recommendation systems that prioritize accuracy employ techniques like collaborative filtering to predict user ratings. We follow this approach, utilizing collaborative filtering to infer the preferences of our customers. For each customer, we compile a list of products ranked by the highest predicted ratings for items they have not yet purchased.

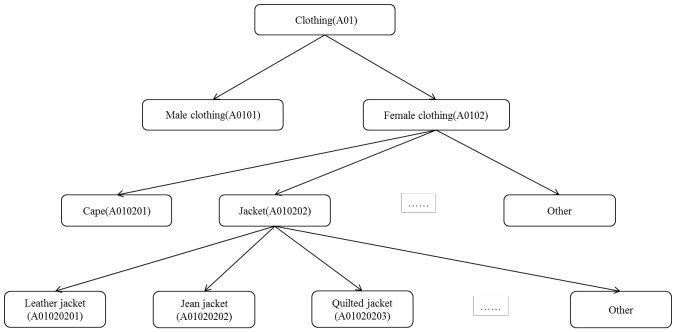


Figure 1: The concept hierarchy of garment products.

To be more specific, for any given customer and their predicted rating r, the list of products recommended based on accuracy is arranged as follows:

$$List_{accuracy} = \{I_1, I_2, \ldots, I_k\} \tag{3}$$

where $r_{I_1} \geq r_{I_2} \geq \ldots \geq r_{I_k}$ and k exceeds the planned number of products for recommendation.

3.1.2 Individual diversity-oriented ranking list. In e-commerce environments, product categories typically exhibit a hierarchical organization. Each product is associated with specific categories, which are structured into a well-defined concept hierarchy. For instance, Figure 1 illustrates a segment of this hierarchy, focusing on various categories within the garment sector.

Within this defined concept hierarchy, we can determine the proximity between two products using a widely accepted metric, which is detailed as follows:

$$dis\left(I_i, I_j\right) = 1 - \frac{2 \times C_3}{C_1 + C_2 + 2 \times C_3} \tag{4}$$

where C_1 and C_2 represent the count of edges from the most specific concepts associated with I_i and I_j up to their nearest shared ancestor, respectively. C_3 denotes the count of edges from this shared ancestor to the hierarchy's root. The ranking list focused on individual preferences is configured as follows:

$$List_{ID} = \left\{ I'_1, I'_2, \ldots, I'_K \right\} \tag{5}$$

where $dis(I'_1, I'_2) \geq dis(I'_i, I'_j)$ for $\forall i, j \leq k$ and $\frac{1}{s-1} \sum_{I'_t \in \{I'_1, I'_2, \ldots, I'_{s-1}\}} dis(I'_s, I'_t) \geq \frac{1}{s-1} \sum_{I'_t \in \{I'_1, I'_2, \ldots, I'_{s-1}\}} dis(I'_{s'}, I'_t)$ for $\forall I'_{s'} \in \{I'_{s+1}, I'_{s+2}, \ldots, I'_k\}$. The positions of I'_1 and I'_2 is established based on their predicted ratings, with the product receiving the higher rating appearing first.

3.1.3 Aggregate diversity-oriented ranking list. Prior research indicates that utilizing a co-purchase network for product recommendations can enhance AD. Within the field, degree centrality measures the sum of connections a node has, which in the context of co-purchasing, reflects how frequently products are bought together. This metric is useful for improving AD. Figure 2 displays a co-purchase network where the numbers adjacent to the links show the frequency of co-purchases between products.

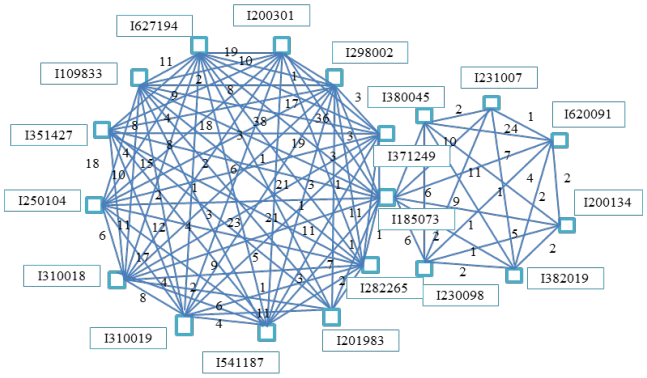


Figure 2: An typical co-purchased network.

The grade of centrality for items within the network of jointly purchased items is determined by the following definition:

$$degree_centrality\,(i) = \sum_{j \in I} f_{i,j} \tag{6}$$

where $f_{i,j}$ denotes the frequency with which product i and product j are co-purchased. We explore the relationship between grade centrality and AD by enhancing the mean grade centrality within recommended product lists and examining the subsequent impact on AD. To further augment AD, we replicate the list-oriented towards accuracy and reorder the products according to ascending degree centrality within the co-purchase network. The resulting list, prioritized for AD, is organized as follows:

$$List_{AD} = \left\{ I_1^{''}, I_2^{''}, \ldots, I_k^{''} \right\} \tag{7}$$

where $degree_centrality(I_i^{''}) \leq degree_centrality(I_{i+1}^{''})$.

3.1.4 Integration of ranking lists. Upon compiling three distinct ranking lists, we enhance both the individual and aggregate diversity of our recommendations by applying a modified version of Borda's count. This method assigns scores to each position within a list and aggregates these scores across the lists to determine the overall ranking of each product. For a given product, the cumulative score from its positions in all three lists yields its final ranking, as determined by this adapted Borda's method:

$$Score_i = \alpha \times Ranking_i^{List_{accuracy}} + \beta \times Ranking_i^{List_{ID}} + \gamma \times Ranking_i^{List_{AD}} \tag{8}$$

where $Ranking_i^{List}$ represents the position of item i within its respective list. The weights α, β, and γ are weight allocated to each of these lists and follows $0 \leq \alpha, \beta, \gamma \leq 1$ and $\alpha + \beta + \gamma = 1$. Following the amalgamation of the three rankings, the composite recommendation list is configured as shown below:

$$Recommendation_list = \left\{ I_1^{'''}, I_2^{'''}, \ldots, I_N^{'''} \right\} \tag{9}$$

where $Score(I_i^{'''}) \leq Score(I_{i+1}^{'''})$.

Finally, the action a_t will take from the *Recommendation_list*. The model we developed operates within the actor-critic framework of reinforcement learning, focusing on the specific roles and functions of actor and the critic network within this structure.

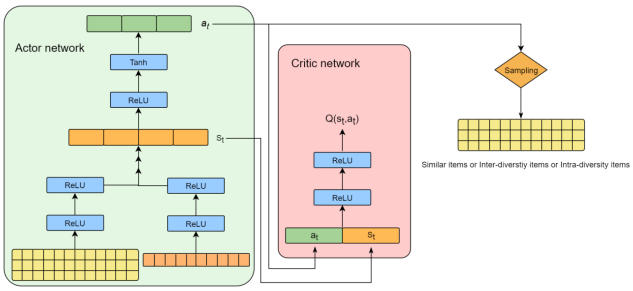


Figure 3: The framework of the proposed D^{3}RDRL approach.

3.2 The actor-network

The actor-network employs a deep learning to determine the most effective strategy, specifically to calculate the optimal action a_t relied on the present state s_t.The initial state s as s_{it}= $\{e_{it} \mid e_{it} \in log_{it}, 0 < i < Length_{rec}\}$, where e_{it} is an embedding of the ith item from the participant's log and the participant's characteristics. When the recommender system produces and presents a set of products to the participant, if the participant overlooks the recommended products, the state of the system does not change, remaining as $s_t = s_{t+1}$. However, if the participant gives positive response to any of the items in D= $\{d_i \mid d_i \in Rec_{rec}, 1 < i < Length_{rec}\}$, which were recommended using methods focused on both accuracy and diversity, like ID&AD re-ranking [26], the system's state is modified accordingly:

$$s_{t+1} = \{\, e_{it} \mid e_{it} \in s_t, r+1 \leq i \leq n \} \cup \{\, d_i \mid d_i \in D, 1 \leq i \leq r \} \tag{10}$$

Our approach to a DRL framework for diverse recommendations is detailed below: At any given state, the actor-network applies DNNs to produce an accomplishment, as depicted in Figure 3. This state is shaped by a user's latest interests, which are derived using a recommendation algorithm. More precisely, the participant's personal attributes are processed through two layers of ReLU-based fully connected layers, generating more refined user features. At each time step, we gather recent items the user has engaged with prior to and input these into another set of two ReLU-based fully connected layers to extract advanced feature levels. The state representation is then formed as $s_t = [u, v_t]$.

Subsequently, the state s_t is processed through subsequent connected layers to produce the action a_t. The initial two layers of the neural network utilize the ReLU activation function, while the output layer employs the tanh activation function to finalize a_t. This process effectively converts the state s into action $a_t = \pi_\theta(S_t)$, which is the output generated by the actor-network.

3.3 The critic network

The trajectory τ produced by the actor guides the critic network, which is tasked with evaluating the Q-value function $Q_\pi(s_t, a_t)$. that judges the action a_t generated by the actor-network matches the current state s_t. After that, based on the $Q(s_t, a_t)$, the actor updates its parameters to improve performance and generate proper actions in the following episode. The critic network employs a deep

Q-network (DQN) with parameter θ^Q, utilizing a deep neural network to approximate the function. Consequently, the Q-function, which predicts the expected returns from a state s_t after taking an action a_t under policy π, is formulated as follows:

$$Q\left(s_t, a_t \middle| \theta^Q\right) = E_{st+1}\left[r_t + \gamma \max_{at+1} Q\left(s_{t+1}, a_{t+1}\right) \middle| \theta^Q\right] \tag{11}$$

3.4 Training

Figure 1 provides a visual representation of the framework we propose. We then explore the methods used for parameter training within this framework. Specifically, we employ the Deep Deterministic Policy Gradient (DDPG) algorithm to adjust and optimize the framework's parameters. The detailed steps of this training process are outlined in Algorithm 1.

Algorithm 1 Model training of D^{3}RDRL.

Arbitrarily set the actor f_{θ^π} and critic network $Q(s, a|\theta^\mu)$ with the constraints $\theta^\pi$and θ^μ.
Set target network f'_{θ^π} and $Q'(s, a|\theta^\mu)$ with utilities $\theta^{\pi'} \leftarrow \theta^\pi$ and $\mu^{\pi'} \leftarrow \mu^\pi$.
Set replay buffer R
for episode←1 to M do
Set state s0 according to the interaction log
for t←1 to T do
Produce action $f_{\theta^\pi}(s_t) = a_t$ based on present policy π
Produce varied list L_t based on Algorithm 1
Suggest list L_t to the consumer
Estimate reward r_t based on Eq. 1
Adjust state from s_t to s_{t+1} using Eq. 1
Record change (s_t, a_t, r_t, s_{t+1}) in$R$.
Set $s_t \leftarrow s_{t+1}$.
Gather a mini-batch of N transitions
(s_i, a_i, r_i, s_{t+1}) from$R$.
Set $y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1}|\theta^{\mu'})|\theta^{Q'})$.
Adjust the Critic network by minimizing the loss:
$loss = \frac{1}{N}\sum_i (y_i - Q(s_i, a_i|\theta^Q))^2$.
Adjust the Actor network with the help of the sampled policy gradient:
$\nabla_{\theta^\mu} J(\theta) \approx \frac{1}{N}\sum_i [\nabla_a Q(s_i, \mu(s_i)|\theta^Q)\nabla_{\theta^\mu}\mu(s_i|\theta^\mu)]$.
Adjust the target networks:
$\theta^{Q'} \leftarrow \tau\theta^Q + (1-\tau)\theta^{Q'}$.
$\theta^{\mu'} \leftarrow \tau\theta^\mu + (1-\tau)\theta^{\mu'}$.
End for
End for

Each cycle of the process is separated into two phases: the transition generation step and the parameter update step. During the transition generation step, starting with the current state s_t, the RA uses Algorithm 1 to suggest a set of items $a_t = \{a_1, a_2, ..., a_k\}$. It then receives a reward r_t from the recommender system and update its state to s_{t+1} accordingly, following the procedure outlined in Algorithm 1. The agent then records the transitions (state, action, reward) in the memory bank D and prepares for the next recommendation by updating the state $s_t = s_{t+1}$. In the subsequent parameter update phase, the RA extracts a mini-batch of transitions (s, a, r, s$'$) from D and adjusts the Actor and Critic constraints using the established DDPG. Our algorithm employs widely recognized techniques such as experience replay and the use of distinct evaluation and target networks to enhance learning consistency and prevent parameter drift. We test our framework in the RecSim NG simulator provided by Google. To evaluate our models, we use both offline and online evaluation methods on datasets available publicly. This dual approach allows us to thoroughly measure the performance and effectiveness of the proposed work in different settings.

4 EXPERIMENTAL EVALUATIONS

4.1 Dataset

Our evaluation of the proposed method involves conducting experimentations with real-world dataset from the Alibaba and Grouplets datasets. Alibaba stands as China's largest e-commerce entity, while the Grouplets datasets are among the most utilized in the field of recommendation systems. To assess the performance, we apply the F-measure (F) for accuracy assessment and also measure individual and aggregate diversities, as detailed in formulas (12, 13, 15, 16) in section 4.2 of our documentation; these metrics are standard in recommendation system evaluations. From the original dataset, which includes over a hundred million interactions between consumers and products, we randomly selected 50,000 transaction records for analysis. Each of these transactions includes a single customer and one or more products, with every product categorized into one or more levels of product hierarchy, from the most general to the most specific. Table 1 delivers a statistical breakdown of the data sampled for our experiments.

4.2 Offline experimental design

For each dataset used in our study, we consider 80% of the user session interactions is considered to be the training set, with the remaining data as the testing set. Specifically, we utilize 40,000 transaction records to adjust and fine-tune the parameters of the user-based collaborative filtering method, which focuses on accuracy. The other 10,000 information are then employed to evaluate the effectiveness of the re-ranked recommendations produced by both the proposed work and other baseline approaches. To assess the precision of these recommendations, we employ a classification accuracy metric in our experimental analysis.

$$\mathrm{F} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

where Precision $= \frac{\#|\mathrm{RS}\cap\mathrm{TS}|}{\#|\mathrm{RS}|}$, Recall $= \frac{\#|\mathrm{RS}\cap\mathrm{TS}|}{\#|\mathrm{TS}|}$, RS denotes the collection of items suggested to clients, while TS symbolizes the test set consisting of items that consumers genuinely prefer. Precision is calculated as the proportion of products in RS that customers find appealing, whereas recall evaluates the percentage of items that have been recommended. F represents the harmonic mean of precision and recall.

The Individual_Diversity metric is employed to assess the ID of the recommendation outcomes. This metric is particularly valuable as it clearly illustrates the range of product diversification within

Table 1: Statistical information of the chosen dataset.

Alibaba Dataset		Movie lens Dataset (1M)	
Data field	Value	Data field	Value
#Transaction	50,000	#Interactions	510,940
#Customer	16,260	#Users	5,218
#Product	95,416	#Items	3,467
#Product category	3,658	#Item category	19

Table 2: The major hyper-parameters.

Method	Major Hyperparameters
ID&AD Re-ranking	$\alpha = 0.4, \beta = \gamma = (1-\alpha)$
DDRLIR	N=$\lvert d_t \rvert$=10, n=$\lvert s_t \rvert$=10, $\lvert Cs_t \rvert$=30, T=[1,10], λ=1.8,d_{model}=100
LIRD	N=$\lvert d_t \rvert$=10, n=$\lvert s_t \rvert$=10, $\lvert Cs_t \rvert$=30, T=[1,10]
DDRL	N= $\lvert d_t \rvert$ =10, n= $\lvert s_t \rvert$ =10, $\lvert Cs_t \rvert$ = 30, T = [1, 10]
Proposed	N= $\lvert d_t \rvert$ =10, n= $\lvert s_t \rvert$ = 3, T = [1, 10]

the recommendations.

$$\text{Individual_Diversity} = \frac{1}{N(N-1)} \sum_{i \in RS} \sum_{j \in RS, j \neq i} \text{dis}(i,j) \tag{13}$$

in which RS represents a list of recommended products, with N indicating the total number of products on that list. The dis(i, j) measures the dissimilarity or conceptual distance between any two products and within the hierarchy, as detailed in Eq.(14) and referenced in [42].

$$\text{dis}\left(I_i, I_j\right) = 1 - \frac{2 \times N_3}{N_1 + N_2 + 2 \times N_3} \tag{14}$$

Here, N_1 and N_2 represent the count of links connecting the most specific perceptions associated with I_i and I_j, respectively, to their nearest shared ancestor, which is the bottom-level concept encompassing both. N_3 denotes the count of links from this shared ancestor to the origin of the concept hierarchy. To assess the AD of the recommended items, we employ two metrics: Gini_diversity and Diversity_in_top_N. These metrics, discussed in detail in section 2.1, highlight distinct yet critical dimensions of AD. They are defined in the following manner:

$$\text{Diversity_in_top_N} = \left| \bigcup_{u \in U} RS_N(u) \right| \tag{15}$$

$$\text{Gini_diversity} = 2 \sum_{i=1}^{|I|} \left[\left(\frac{|I| + 1 - i}{|I| + 1} \right) \times \left(\frac{rec_i}{total} \right) \right] \tag{16}$$

Here, N represents the count of ietms suggested to an individual consumer, $RS_N(u)$ denotes the collection of items suggested to consumer u, i indexes a product within set I, rec_i quantifies how many customers receive recommendations for product i. The products I are sorted by rec_i in increasing order, and total reflects the aggregate count of product suggestions made to all consumers. To predict product preferences for a given customer, we employ user-based CF. Products are then ranked according to these preferences, with the top NNN products recommended to the customer, using what is referred to in our study as the conventional ranking method [23], which ranks items in non-increasing order based on predicted favorites. Beyond conventional methods, our research also explores reinforcement learning techniques to enhance the recommendation process and compare with our proposed model's performance, explicitly employing LIRD, DDRL, and DDRLIR, noted for incorporating diversity and serving as standard baselines. Key hyper-parameters of these methods are detailed in Table 2.

4.3 Offline experimental outcomes and algorithm estimation

Figures 4 and 5 present a summary of the evaluation results. These results indicate that the proposed approach beats competing systems in achieving higher diversity and precision across both the MovieLens and Alibaba datasets.

4.4 Online experimental design

In an offline environment, interactive recommender systems face limitations due to static user feedback, which hampers their ability to adapt to changing user preferences. Following the approach outlined in [43], we address this challenge by employing the RecSim NG simulator to conduct simulated online experiments using the Alibaba Dataset and Movielens-1M datasets. For these online evaluations, we develop a simulator to benchmark our method against DDRLIR in an interactive setting, as detailed in Algorithm 1. Our simulated experiments reveal that both our method and DDRLIR experience fluctuations in online recommendation accuracy across various epochs as depicted in Figure 6(a). However, our method consistently surpasses DDRLIR in terms of accuracy at every epoch. Figures 6(b) and 6(c) further illustrate that our recommendations exhibit significantly higher levels of both individual and aggregate diversity compared to those by DDRLIR. Specifically, our method shows an improvement of 3.2%, 1.2%, and 6.3% in precision, individual diversity, and aggregate diversity, respectively, across all epochs in the Movielens-1M dataset. Additionally, the performance on the Alibaba Dataset, as shown in Figure 7, reinforces these findings. Both methods achieve stable accuracy levels, with our approach

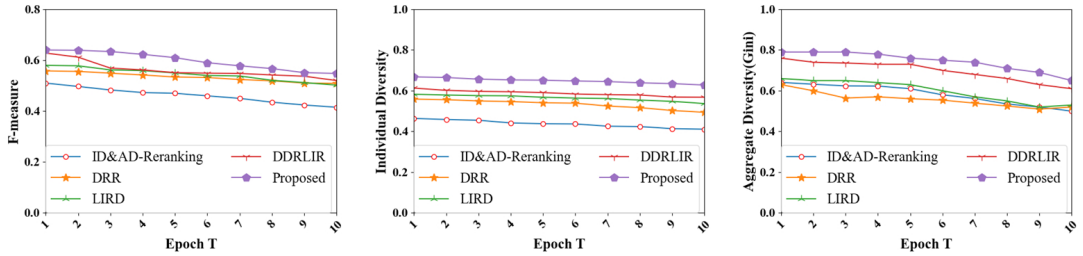


Figure 4: Results of offline assessments using the MovieLens-1M dataset.

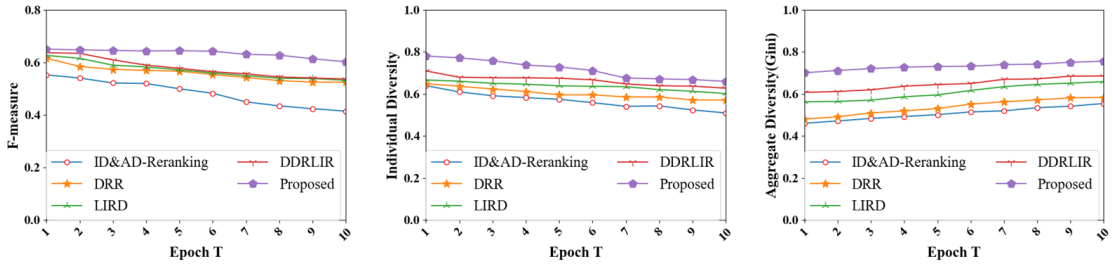
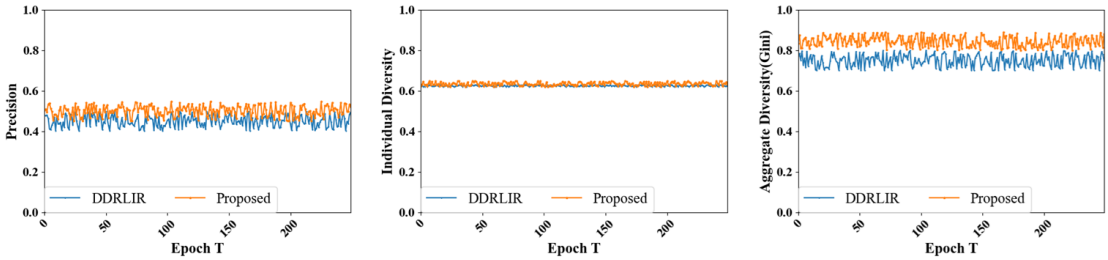


Figure 5: Results of offline assessments using the Alibaba-Dataset.



(a) (b) (c)

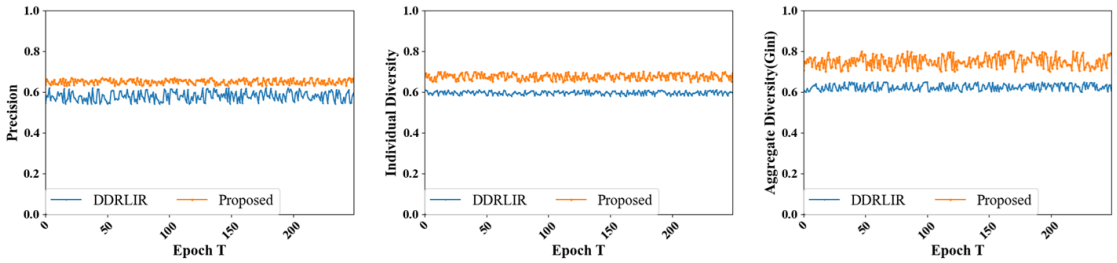
Figure 6: Results of online assessments using the MovieLens-1M.

consistently outperforming DDRLIR across all recommendation epochs as shown in Figure 7(a). Moreover, the recommendations made by our method demonstrate considerably greater individual and aggregate diversity, outperforming DDRLIR by 7.5% and 8.1% respectively, as illustrated in Figures 7(b) and 7(c). These outcomes underscore the higher performance of our approach in adapting to user preferences dynamically in simulated environments.

5 CONCLUSIONS

In conclusion, we introduce a DRL framework tailored for an interactive recommendation system that optimizes both the relevance and diversity of its suggestions, dubbed **D^{3}RDRL**. The framework executes three distinct actions to generate ranking lists focused on accuracy, ID, and AD, enhancing both aspects while retaining precision. We validate our approach through tests on a real-world dataset, where our method shows notable enhancements in both recommendation precision and diversity. Future studies will explore the adaptability of this technique in various recommendation scenarios. This approach holds promise for e-commerce environments, aiding them in achieving a fine-tuned balance between accurate and diverse recommendations. Moreover, the strategy could be extended to other recommendation domains, such as suggesting academic collaborators across different disciplines or diverse citation recommendations for scholarly writing. In academic collaborations, for example, a researcher might seek out experts from varied fields, while in scholarly article writing, diverse citation recommendations can cover broader topics within the paper. For enhancing diversity, a research area classification tree could guarantee ID, complemented by a citation network to uphold AD.

In future work, we aim to further assess the scalability of our method across diverse recommendation settings and are keen on crafting algorithms that introduce temporal diversity using deep reinforcement learning. Additionally, a significant challenge in Top-N recommendation tasks involves managing a vast item space,



(a) (b) (c)

Figure 7: Results of online assessments using the Alibaba-Dataset.

potentially running into millions. To address this, we recommend deploying a distributed system that simplifies the complex combinatorial recommendation challenge by breaking it down into manageable, item-level recommendations.

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